UCI Diabetes 130-US hospitals

Preliminary EDA

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Loading of data

```
library(readr)
library(here)
```

```
## here() starts at /Users/ben/Library/CloudStorage/OneDrive-Persönlich/Ben/TU Wien/Semester5/compstat/
```

```
file_path <- here("data", "raw", "diabetic_data.csv")


raw_df <- read_csv(file_path,
                   na = c("?", "NA", ""),
                   show_col_types = FALSE)
```

Handling of "Not Available" data

```
#percentage of NAs for every column
na_percent <- colMeans(is.na(raw_df)) * 100
sort(na_percent[na_percent > 0], decreasing = TRUE)
```

```
##            weight medical_specialty        payer_code              race
##       96.85847926       49.08220820       39.55741603        2.23355541
##            diag_3            diag_2            diag_1
##        1.39830592        0.35178743        0.02063558
```

Variable	Missingness	Strategy	Methodological Rationale
weight	>96%	Column Omission	Unusable feature; imputation introduces noise.
payer_code	~40%	Column Omission	Administrative billing category, not clinical.
medical_specialty	~49%	Recode to "Missing"	Informative missingness (proxy for ER admission).

Variable	Missingness	Strategy	Methodological Rationale
race	~2%	Row Omission	Complete-case filtering; preserves >97% of sample.
diag_1, diag_2, diag_3	<1%	Row Omission	Complete-case filtering on essential clinical target.